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Ocean Target Electric Field Signal Analysis and Detection Using LOFAR Based on Basis Pursuit

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Abstract: An ocean target electric field signal is an effective approach for analyzing the ocean environment and is widely used for detecting ocean targets, extracting their features, and tracking them. Low-frequency analysis and recording (LOFAR) is a commonly used time–frequency analysis tool that provides the time–frequency spectrum of a signal; however, its reliance on the Fourier transform (FT) results in a low frequency resolution and signal-to-noise ratio (SNR), which limits its target detection capabilities. To address this problem, we propose a method called low-frequency analysis and recording based on basis pursuit (LOFAR-BP) for analyzing and detecting ocean target electric field signals. LOFAR-BP uses basis pursuit (BP) with the L1 norm for frequency analysis, whereas LOFAR utilizes the FT. We demonstrate that the FT is the L2 norm mathematically. LOFAR-BP generates the time–frequency spectrum in the same way that LOFAR does. By extracting characteristic values from the time–frequency spectrum, targets can be detected using an appropriate threshold. Both simulation and ocean experiments showed that LOFAR-BP effectively enhances target signals and suppresses noise. Compared with LOFAR, LOFAR-BP improved the frequency resolution by 60% in both experiments and increased the SNR by 54.82 dB in the simulation experiment and by 39.59 dB in the ocean experiment. When applied to target detection, LOFAR-BP can detect targets 6 s earlier than LOFAR can.

Keywords: low-frequency analysis and recording; basis pursuit; electric field signal; time–frequency spectrum; target detection



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1. Introduction

In the ocean, when an ocean target travels, it generates not only sound signals but also electric field signals. In the modulation of its main shaft and propeller, the corrosion and anticorrosion current produce a shaft-rate electric field signal. This electric field signal is a multiplication frequency signal with the fundamental frequency corresponding to the rotation frequency of the propeller. Owing to its distinct line spectrum, electric field signal analysis is crucial for analyzing the ocean environment [1].

The Ag-AgCl sensor widely used records the electric field signal. It is often equipped with an electric field measurement system underwater in a measurement ship. Some studies adopt a single long detection line consisting of a pair of current electrodes and multiple potential electrodes to record the electric field signal [2]. Analyzing the electric field signal generated by other ocean targets, we can obtain the time–frequency spectrum and extract the ocean target characteristics. From the ocean target characteristics, using

energy detection, tracking line spectrum technology, and convolutional neural network (CNN), we can detect, track, and recognize the ocean target.

The Fourier transform (FT) is commonly used to obtain the frequency spectrum of an electric field signal [2]. In the frequency spectrum, a sharp and high peak often appears when a target is present. However, owing to the strong noise and weak electric field signals, the line spectrum can be obscured in the frequency spectrum. To address this problem, researchers have used the adaptive line enhancer (ALE) to preprocess the electric field signal [3,4], which improves the signal-to-noise ratio (SNR). Nevertheless, the FT is a tool that analyzes signals only in the frequency domain. To analyze a signal simultaneously in both the time domain and frequency domain, the short-time Fourier transform (STFT) was developed.

Low-frequency analysis and recording (LOFAR), which is similar to the STFT, is widely used in ocean target detection, tracking, classification, etc. [5–12], owing to its effectiveness in the time-low-frequency domain where ocean targets typically exist. Silva et al. [5] proposed an automatic detection and tracking method using LOFAR, which was validated by processing a real dataset from a flank array in a navigation channel. Additionally, Wang et al. [6] introduced a passive tracking method for underwater acoustic targets using LOFAR. The results demonstrated the effectiveness of the method for tracking underwater acoustic targets with intersecting azimuth trajectories. Luo et al. [13] proposed a one-dimensional hidden Markov model (1D-HMM) method to extract the time-varying line spectrum of underwater acoustic targets using LOFAR. Their results showed that this method can effectively detect and track the frequency states in the time–frequency spectrum obtained using LOFAR, even at a low SNR. Park and Jung [14] combined LOFAR and CNN to investigate 175 CNN models for identifying the tonal frequency of acoustic targets. Their results revealed that the maximum accuracy of the best model was 96.2% for precision and 99.5% for recall. Likewise, Li et al. [15] proposed a method for the simultaneous detection of multiple underwater acoustic targets. An analysis of a lake trial dataset revealed that this method achieved an accuracy of 98%, outperforming traditional CNN methods. These advancements highlight the broad applicability of LOFAR in ocean target detection and related fields.

Similar to LOFAR, the wavelet transform (WT) is also a time–frequency analysis method based on the principle of orthogonal decomposition. WT achieves the best balance between time resolution and frequency resolution. WT can filter signals, detect signals, and extract signal time–frequency characteristics [2,16,17]. Kim et al. [2] analyzed an electric field signal to detect ocean targets in real-time using the WT design method - the multiresolution analysis (MRA) - and a field experiment verified its effectiveness. By defining a type of function, namely, the intrinsic mode function (IMF), the empirical mode decomposition (EMD) decomposes a signal into many IMFs, where each IMF represents a frequency signal. EMD has a very low computation cost. Fan et al. [18] proposed an improved EMD method that can effectively suppress the endpoint effect, and the electric field signal analysis revealed that the method improved the system's lightning location capabilities. By solving a constrained programming problem that involves an overcomplete dictionary and the L1 norm as the optimization function, basis pursuit (BP) can analyze a signal [19–24]. By setting the overcomplete dictionary as a Fourier dictionary, Lang et al. [22] used a method that is based on BP to filter gravity data. The results showed that the method reduced edge effects while maintaining the accuracy of the finite impulse response (FIR). Gao et al. [23] also applied this method to predict ocean tides. They reported that the method achieved a higher prediction accuracy than harmonic analysis (HA) did when data from the Hong Kong station were processed. Furthermore, Liu et al. [24] used a method that is based on BP to detect speech signals. Their results revealed that this method

was more robust in a low-SNR environment. These studies contribute to the wide usage of BP in signal analysis.

Notably, LOFAR has a low frequency resolution because of its reliance on its basic step, the FT, which has limited frequency resolution [25,26]. The limitation further hinders its ability to extract frequency characteristics. To improve its frequency resolution, the combination of LOFARA and BP, which is referred to as low-frequency analysis and recording based on basis pursuit (LOFAR-BP), is proposed in this paper. LOFAR-BP uses an overcomplete Fourier dictionary, which makes the frequency interval arbitrarily narrow to potentially improve the frequency resolution. Additionally, LOFAR-BP adopts the BP to constrain the frequency spectrum, which makes the frequency spectrum sparse. LOFAR-BP achieves high frequency resolution and can highlight target signals and suppress noise, which improves the ability to analyze, detect, and track ocean targets.

This paper proposed a new time–frequency analysis method named LOFAR-BP. LOFAR-BP can obtain the exact time–frequency characteristics of a target signal with a higher frequency resolution. Using the exact time–frequency characteristics, LOFAR-BP can detect the ocean target earlier in a lower SNR ocean experiment. The remainder of this paper is organized as follows: In Section 2, LOFAR and BP are reviewed briefly, and LOFAR-BP and target detection are described in detail. In Section 3, a simulation experiment is described, and an analysis using LOFAR and LOFAR-BP is presented. In Section 4, an ocean experiment is conducted, and the analysis and detection of ocean target electric field signals using LOFAR and LOFAR-BP are presented. In Section 5, the conclusions and discussion are presented. In Appendix A, theorems used in this paper are proven.

2. LOFAR-BP

LOFAR is a time–frequency analysis tool that divides a signal into many overlapping segments, which allows it to analyze the signal in the time domain. For each segment, it uses a frequency analysis tool, the FT, to analyze the signal in the frequency domain. Owing to the main two steps, LOFAR can obtain the time–frequency spectrum of the signal, which is referred to as the LOFAR spectrum. If the method of dividing the signal is fixed, different frequency analysis tools result in different kinds of LOFARs. In this paper, a frequency analysis tool that is based on the BP is constructed, and the relationship between it and the FT is demonstrated. LOFAR-BP replaces the FT with the frequency analysis tool. LOFAR-BP can also obtain the time–frequency spectrum of the signal, which is referred to as the LOFAR-BP spectrum. Owing to the sparsity of the frequency analysis tool, LOFAR-BP has better properties in the time–frequency spectrum than LOFAR does.

2.1. LOFAR

Let a signal be $X[n]$, $0 \leq n \leq N - 1$. The signal is divided using the following formula:

$$X_i[n] = X[n + iS] \quad 0 \leq i \leq i_{max}, \quad 0 \leq n \leq L - 1 \quad (1)$$

where L denotes the window width; S denotes the step length; i denotes the segment number; i_{max} denotes the maximum of i , which is the largest integer no greater than $(N - L)/S$; and X_i denotes the i th segment of X .

Before analyzing each segment, preprocessing for normalization and demeaning should be implemented as follows:

$$\begin{aligned} X_i[n] &= X_i[n] / \max(|X_i|) \\ X_i[n] &= X_i[n] - \bar{X}_i \end{aligned} \quad (2)$$

where $|\cdot|$ denotes the absolute value operator; $\max$ denotes the maximum function; and $\bar{\cdot}$ denotes the average operator.

After preprocessing, the FT can be used to obtain the frequency spectrum. Let the FT result of X_i be H_i^{FT} .

$$H_i^{FT}[m] = \sum_{n=0}^{n=L-1} X_i[n] \exp\left(j(-1)^m \frac{2\pi}{L} n\right) \quad 0 \leq m \leq L-1 \quad (3)$$

Through division (Formula (1), time domain analysis) and the FT (Formula (3), frequency domain analysis), the LOFAR spectrum of the signal is obtained as follows:

$$Y^{FT}[n, m] = H_i^{FT}[m] \quad 0 \leq i \leq i_{max}, n = iS + (L-1)/2 \quad (4)$$

where n denotes the time, the middle time of the i st segment of the signal; m denotes the frequency; and Y^{FT} denotes the LOFAR spectrum.

In the FT, the frequency interval is $1/L$. To obtain a smaller frequency interval, the signal is usually extended by adding zeros. However, this approach does not improve the frequency resolution [25], which is not conducive to the LOFAR spectrum and further target detection. The steps of LOFAR are shown in Figure 1.

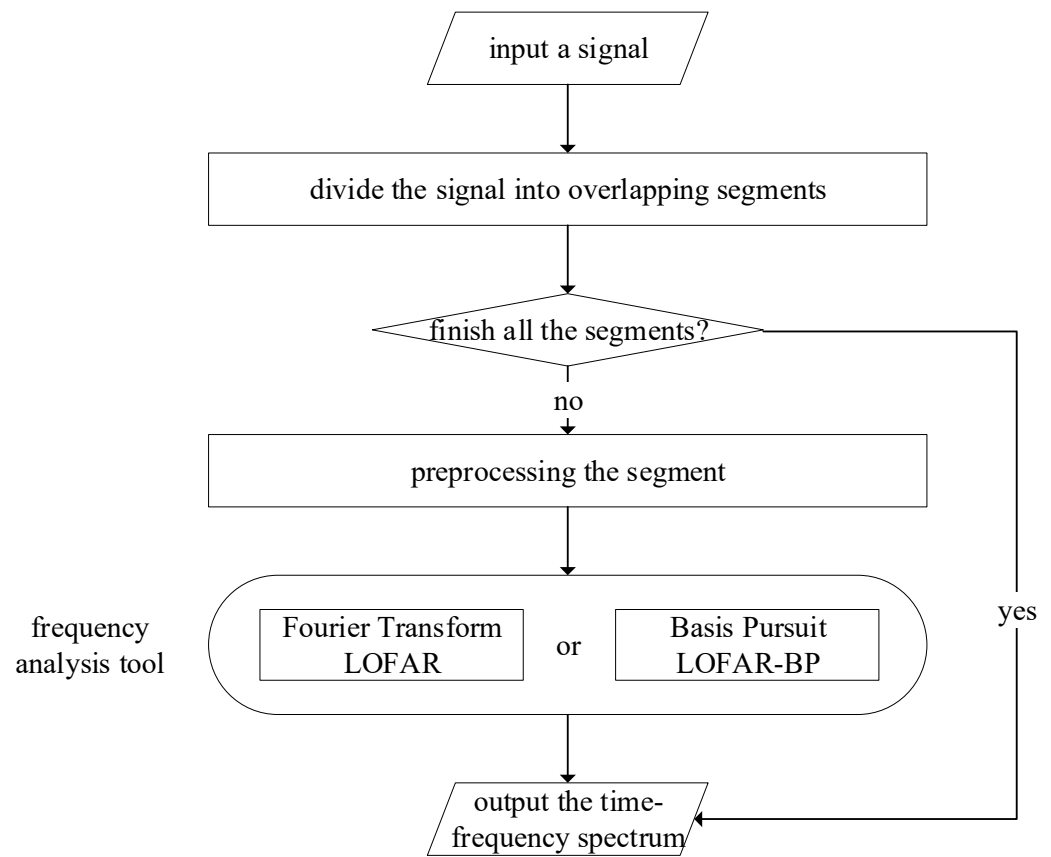


Figure 1. Steps of LOFAR and LOFAR-BP.

2.2. Basis Pursuit

BP is a constrained programming problem [27], as shown below:

$$\begin{aligned} \min \quad & \|x\|_1 \\ \text{s.t.} \quad & \Phi x = y \end{aligned} \quad (5)$$

where Φ denotes the dictionary, an $N \times K$ matrix; y denotes the signal, an $N \times 1$ column vector; x denotes the expression, a $K \times 1$ column vector; and $\|\cdot\|_1$ denotes the L1 norm operator.

In BP, each column vector in Φ is referred to as an atom, and K is the number of atoms. Because Φ is an overcomplete dictionary, which indicates that K is much larger than N , there are many solutions for $\Phi x = y$. The L1 norm as an optimization function makes the optimal solution x^* sparse, which indicates that most components of x^* are close to 0.

Although BP is a linear problem that can be solved, the time cost for obtaining the global optimal solution is too large when a large amount of data is used, which is not practical for real-world applications. In this paper, the algorithm smoothed L1 norm (SL1) [28] is programmed to solve BP. SL1 obtains a suboptimal solution in a limited time and number of iterations, which is sufficient for analysis.

2.3. LOFAR-BP

In BP, Φ is set as shown below:

$$\Phi = \frac{1}{K} \begin{pmatrix} \exp(j0\frac{2\pi}{K}0) & \exp(j1\frac{2\pi}{K}0) & \dots & \exp(j(K-1)\frac{2\pi}{K}0) \\ \exp(j0\frac{2\pi}{K}1) & \exp(j1\frac{2\pi}{K}1) & \dots & \exp(j(K-1)\frac{2\pi}{K}1) \\ \dots & \dots & \dots & \dots \\ \exp(j0\frac{2\pi}{K}(N-1)) & \exp(j1\frac{2\pi}{K}(N-1)) & \dots & \exp(j(K-1)\frac{2\pi}{K}(N-1)) \end{pmatrix} \quad (6)$$

where K is the extended length, which indicates that the addition of zeros behind the signal increases the signal length from N to K .

If the objective function is an L2 norm, Formula (5) is rewritten as follows.

$$\begin{aligned} \min \quad & \|x\|_2 \\ \text{s.t.} \quad & \Phi x = y \end{aligned} \quad (7)$$

Theorem 1 (details are in Section Appendix A) proves that the optimal solution for Formula (7) is the same as the FT result when the signal y is extended into the length K by adding zeros. The frequency spectrum of a signal can be obtained by solving a constrained programming problem. Similarly, keeping the equality constraint and restoring the objective function back to the L1 norm as BP does, a frequency analysis tool that is based on BP is constructed.

For each segment X_i described in Section 2.1 LOFAR, the frequency spectrum of segment X_i obtained using the frequency analysis tool that is based on BP is denoted by H_i^{BP} , which is the optimal solution for Formula (5), where Φ and y are Formula (6) and segment X_i , respectively. Then, the LOFAR-BP spectrum of the signal X is obtained as follows:

$$Y^{BP}[n, m] = H_i^{BP}[m] \quad 0 \leq i \leq i_{max}, n = iS + (L-1)/2 \quad (8)$$

where n denotes the time, the middle time of the i th segment of the signal; m denotes the frequency; and Y^{BP} denotes the LOFAR-BP spectrum.

The essential difference between LOFAR and LOFAR-BP is the frequency analysis tool. The LOFAR uses the FT, whereas LOFAR-BP uses the frequency analysis tool that is based on BP. The steps of LOFAR and LOFAR-BP are shown in Figure 1.

2.4. Target Detection

As a conventional method for target detection, energy detection uses the energy of a signal as the characteristic value of the signal. In this work, we defined the characteristic value as the maximum of the absolute value of the frequency spectrum. Similarly, for a time–frequency spectrum, the characteristic value varies over time, and the calculation formula is as follows:

$$\lambda[n] = \max_{m \in Q} |Y[n, m]| \quad (9)$$

where Y denotes the time–frequency spectrum; Q denotes the frequency band; and λ denotes the characteristic value.

Target detection involves comparing the characteristic value with a threshold. In this work, we adopted an easy way to calculate the threshold, which is the average of all characteristic values. If the characteristic value exceeds the threshold, the detection result is set to true; if the characteristic value is below the threshold, the detection result is set to false.

3. Simulation Experiment

We designed a simulation experiment to test the performance of LOFAR-BP. In the simulation experiment, a signal (a mathematical signal without a unit) is simulated as a chirp signal with noise. We used both LOFAR and LOFAR-BP to analyze the signal and obtain its time–frequency spectrum. When the two time–frequency spectra are compared, several conclusions are drawn.

3.1. Simulated Signal

In the simulation experiment, the simulated signal is constructed using the following formula:

$$f(t) = \cos\left(2\pi\left(\frac{1}{2000}t^2 + \frac{3}{5}t\right)\right) + N(0, 0.1^2) \quad t \in (0, 900) \quad (10)$$

where $N(\mu, \sigma^2)$ denotes the Gaussian distribution with the expectation μ and the variance σ^2 .

Equation (10) shows that the simulated signal consists of a chirp signal and white Gaussian noise. The frequency of the chirp signal changes from 0.6 Hz at 0 s to 1.5 Hz at 900 s. The expectation and variance of the white Gaussian noise are 0 and 0.1^2 , respectively. The simulated signal is shown in Figure 2.

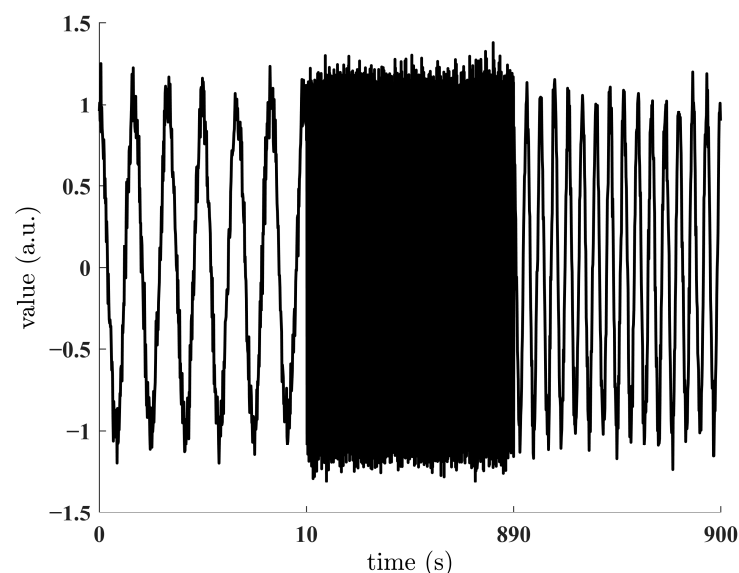


Figure 2. Simulated signal.

In Figure 2, the x -axis is nonuniform globally but uniform between two narrow x -axis ticks. Focusing on $t \in (0, 10)$ and $t \in (890, 900)$, the frequency clearly increases. Additionally, the sampling frequency is designed to be 50 Hz.

3.2. Time–Frequency Spectrum Analysis

The simulated signal was processed using LOFAR and LOFAR-BP. The window width L , step length S , and extended length K are 1001, 450, and 6006, respectively. Considering the sampling frequency of 50 Hz, the window width, step length, and frequency interval are 20.02 s, 9 s, and 0.0083 Hz, respectively. The two time–frequency spectra obtained by LOFAR and LOFAR-BP are shown in Figure 3.

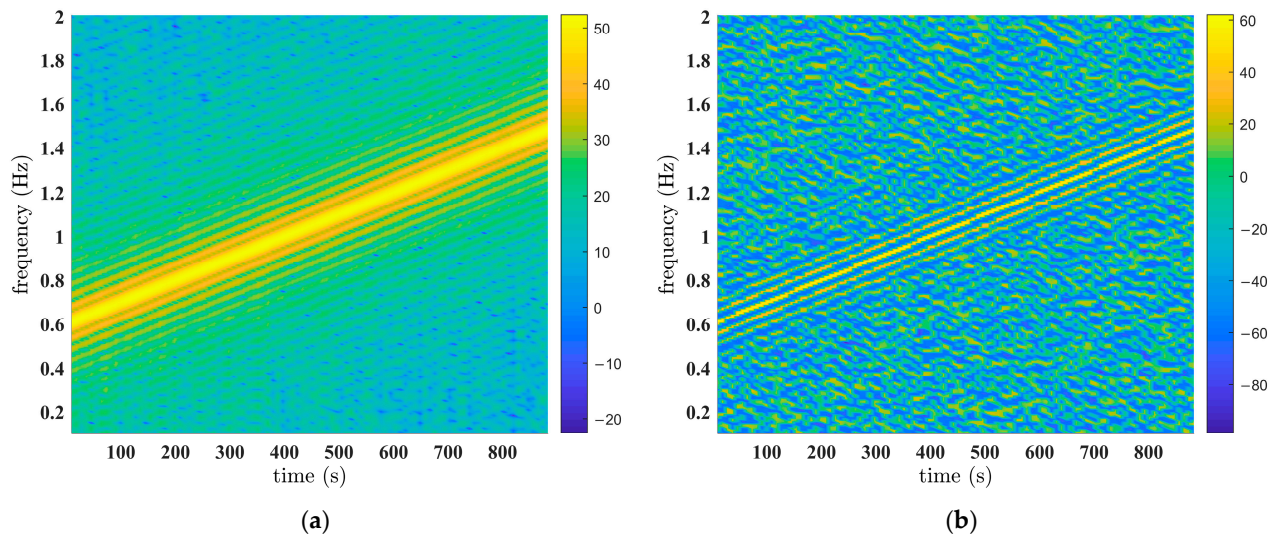


Figure 3. Time–frequency spectra of simulated signal. (a) LOFAR. (b) LOFAR-BP.

In Figure 3, the distinct yellow line is a target signal - the chirp signal - in the simulated signal. Outside the distinct yellow line, Figure 3a shows many lines parallel to the distinct yellow line. However, Figure 3b shows many irregular spots of noise. This finding indicates that LOFAR-BP collects the energy of the target signal in a small time–frequency area. Additionally, the distinct yellow line in Figure 3a is thick, whereas the line in Figure 3b is thin, which indicates that the frequency resolution of LOFAR-BP is higher than that of LOFAR. To compare the performance of LOFAR and LOFAR-BP in the frequency spectrum, two frequency spectra were extracted from the respective time–frequency spectra at 600 s, as shown in Figure 4.

Figure 4 shows that the energy range of LOFAR-BP (−74.23 to 60.31 dB) is larger than that of LOFAR (−7.08 to 51.72 dB), indicating that LOFAR-BP highlights the target signal into a higher energy and suppresses the noise into a lower energy. The LOFAR-BP peak is higher and sharper than the LOFAR peak; thus, LOFAR-BP not only enhances the target signal but also has a higher frequency resolution. To evaluate the frequency resolution numerically, we defined the peak width in the frequency spectrum as the frequency resolution. Looping through all the moments, the frequency resolutions varying over time of LOFAR and LOFAR-BP are shown in Figure 5.

Figure 5 shows that LOFAR-BP has a higher frequency resolution than LOFAR does. The average frequency resolutions of LOFAR and LOFAR-BP are 0.1 Hz and 0.04 Hz, respectively. These results indicate that, compared with LOFAR, LOFAR-BP improves the frequency resolution by 60%.

In Figure 3, regarding the distinct yellow line and the residual as the target signal and noise, separately, several energy statistics were calculated, as shown in Table 1. Figure 4.

In Table 1, the values of the minimum energy indicate that LOFAR-BP has a better ability to suppress the noise, and the values of the maximum energy indicate that LOFAR-BP has a better ability to highlight the target signal. The values of the SNR indicate that, compared with LOFAR, LOFAR-BP improves the SNR by 54.82 dB.

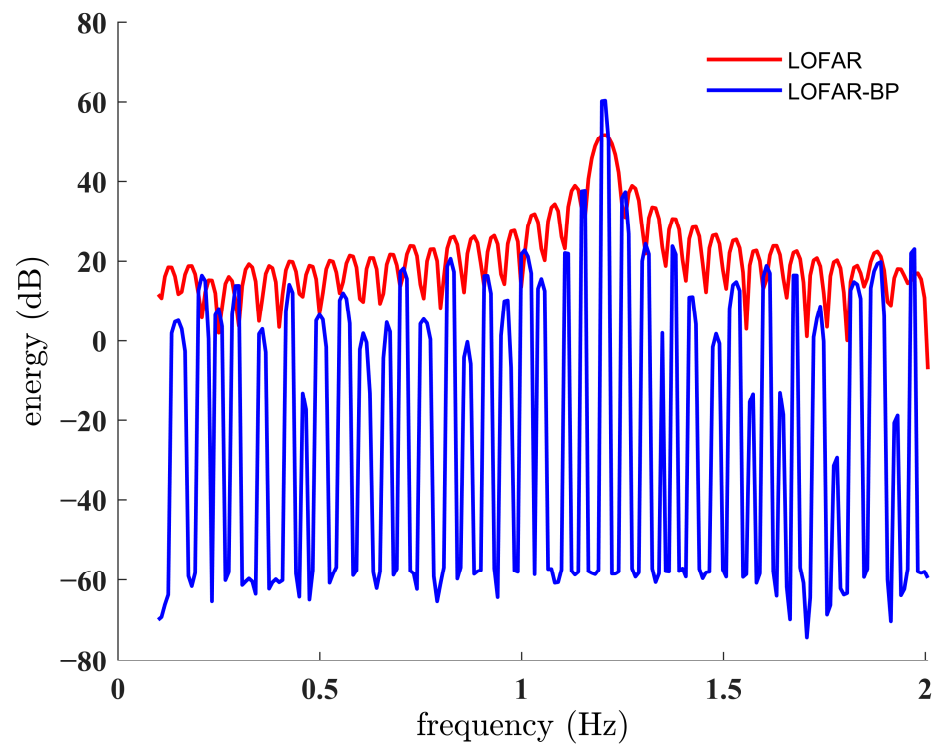


Figure 4. Frequency spectrum.

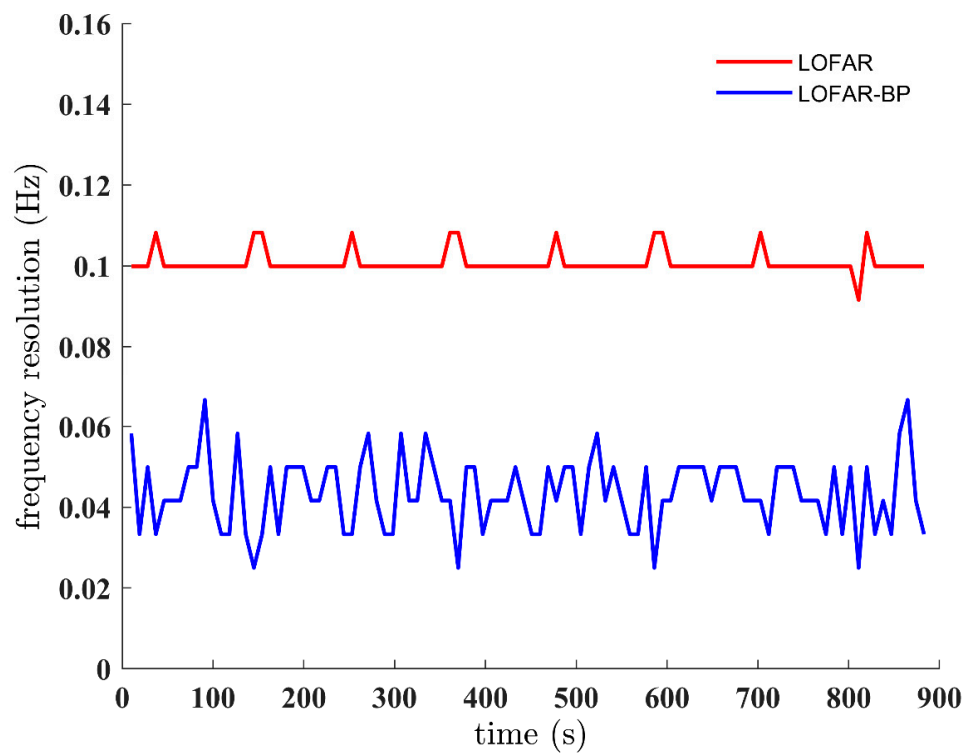


Figure 5. Frequency resolution of simulated signal.

Table 1. Energy statistics about simulated signal.

Method	Minimum Energy (dB)	Maximum Energy (dB)	SNR (dB)
LOFAR	−22.52	52.41	31.49
LOFAR-BP	−98.37	62.08	86.31

4. Ocean Experiment

To verify the superiority of LOFAR-BP in a real environment, an ocean experiment was performed. In the ocean experiment, the electric field signal was recorded using an Ag-AgCl sensor equipped with an electric field measurement system underwater. While the electric field signal was recorded, a ship passed by. The ship was far from the measurement system at the start and end but close to the measurement system at the middle of the ocean experiment. During the travel, the Ag-AgCl sensor in the measurement system recorded the electric field signal continuously.

4.1. Ocean Target Electric Field Signal

The electric field signal was sampled at a sampling frequency of 50 Hz, as shown in Figure 6.

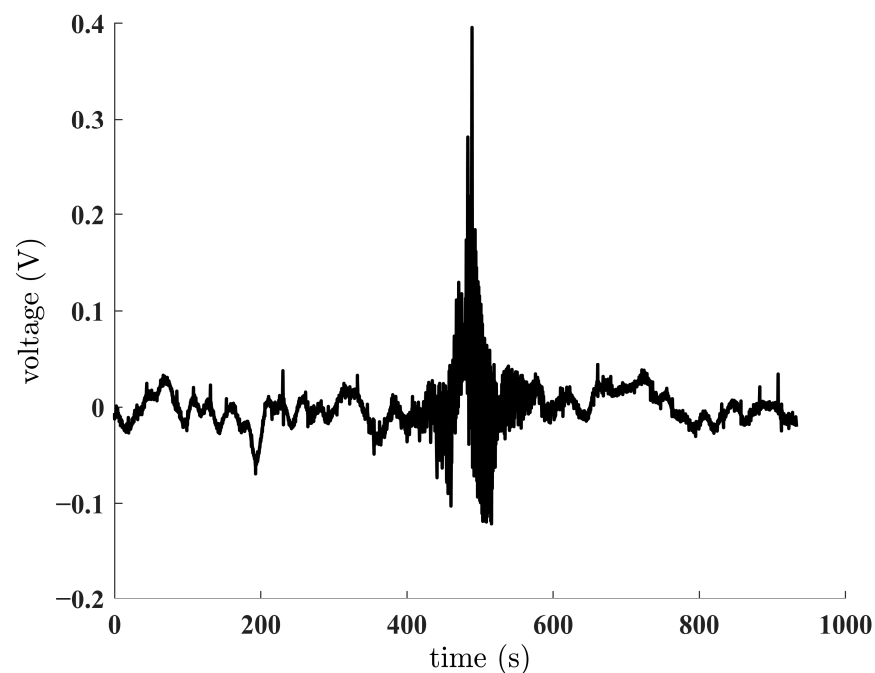


Figure 6. Electric field signal.

In Figure 6, the voltage of the electric signal always remains low, with the exception of $t \in (400, 600)$, which meets the real circumstance that the distance between the ship and the measurement system was far at the start and end but close at the middle of the ocean experiment.

4.2. Time–Frequency Spectrum Analysis

The electric field signal was processed using LOFAR and LOFAR-BP. The window width L , step length S , and extended length K are 2001, 25, and 12,006, respectively. Considering the sampling frequency of 50 Hz, the window width, step length, and frequency interval are 40.02 s, 0.5 s and 0.0042 Hz, respectively. The two time–frequency spectra obtained by LOFAR and LOFAR-BP are shown in Figure 7.

Both Figure 7a,b show that the electric field signal contains a single-frequency signal with a frequency of 1 Hz and duration of $t \in (300, 700)$. However, the target signal, which is the single-frequency signal, is thinner in Figure 7b than that in Figure 7a. For convenience, we truncated the time–frequency area to $t \in (300, 700) \times f \in (0.6, 1.5)$ in which both the target signal and noise exist. The peak width in the frequency spectrum is defined as the frequency resolution, the same as the simulation experiment does. Looping through all the

moments, the frequency resolutions varying over time obtained by LOFAR and LOFAR-BP are shown in Figure 8.

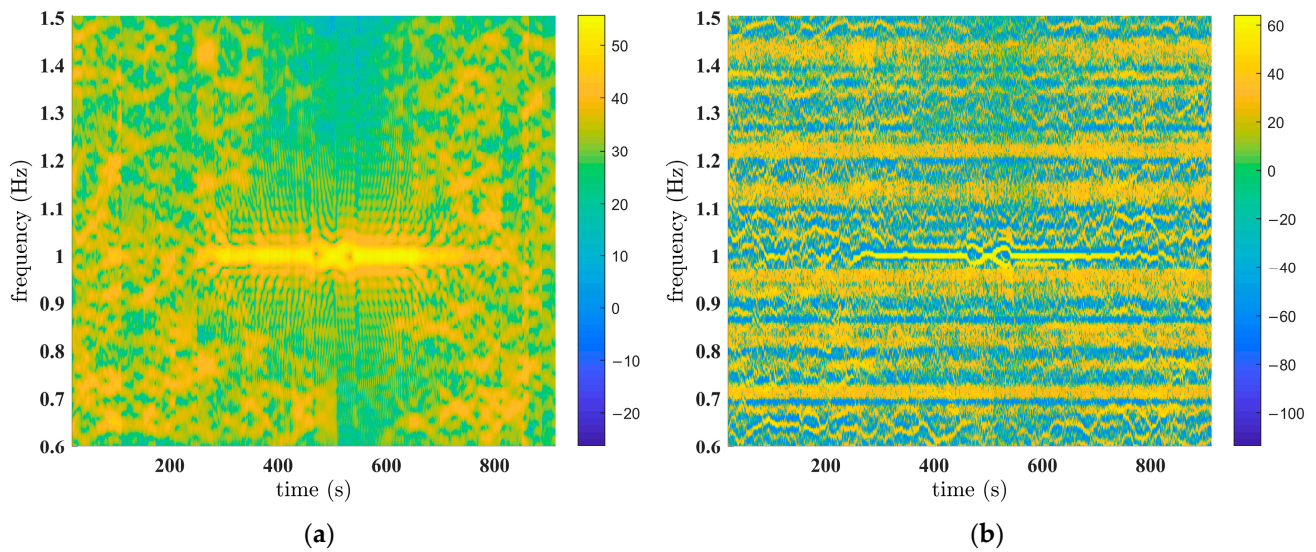


Figure 7. Time–frequency spectra of electric field signal. (a) LOFAR. (b) LOFAR-BP.

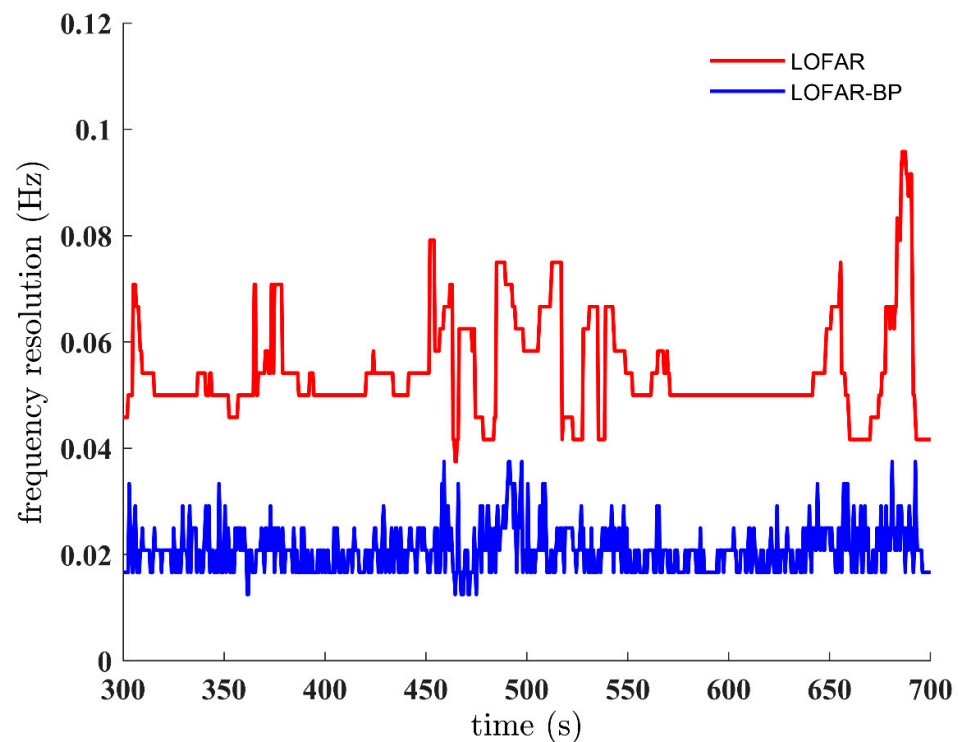


Figure 8. Frequency resolution of electric field signal.

Figure 8 shows that LOFAR-BP has a higher frequency resolution than LOFAR does. The average frequency resolutions of LOFAR and LOFAR-BP are 0.05 Hz and 0.02 Hz, respectively; therefore, compared with LOFAR, LOFAR-BP improves the frequency resolution by 60%. In the truncated time–frequency area $t \in (300, 700) \times f \in (0.6, 1.5)$, regarding the single-frequency signal and the residual as the target signal and noise, separately, several energy statistics were calculated, as shown in Table 2.

Table 2. Energy statistics about electric field signal.

Method	Minimum Energy (dB)	Maximum Energy (dB)	SNR (dB)
LOFAR	−26.21	55.85	22.54
LOFAR-BP	−109.11	64.04	62.13

In Table 2, LOFAR-BP has a smaller minimum energy value and a larger maximum energy value than LOFAR does, indicating that LOFAR-BP performs better than LOFAR does in both suppressing the noise and highlighting the target signal. Moreover, compared with LOFAR, LOFAR-BP improves the SNR by 39.59 dB.

4.3. Target Detection

In Formula (9), the time–frequency spectrum Y is set to the LOFAR spectrum or LOFAR-BP spectrum, and the frequency band Q is set to $f \in (0.6, 1.5)$. We calculated the characteristic value. The characteristic values of LOFAR and LOFAR-BP are shown in Figure 9.

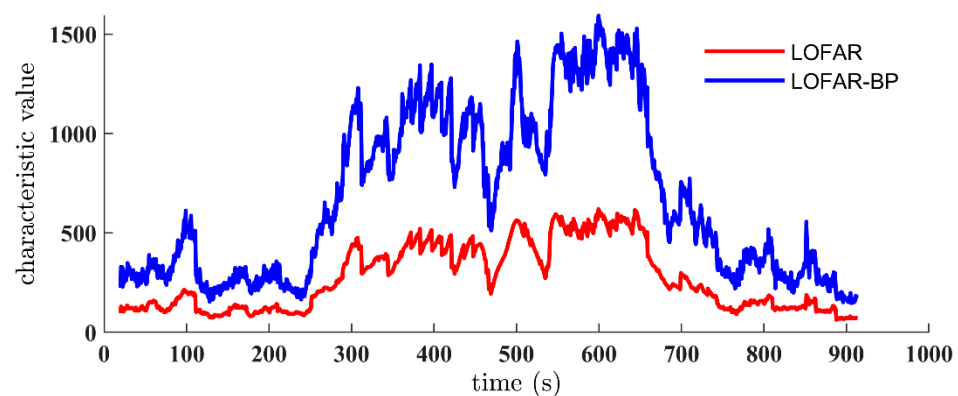
**Figure 9.** Characteristic value.

Figure 9 shows that the characteristic value of LOFAR-BP is always larger than that of LOFAR. The difference is small for $t \in (0, 200)$ and $t \in (750, 900)$, whereas it is large for $t \in (300, 650)$. For both LOFAR and LOFAR-BP, the threshold is set to the average of the respective characteristic values. The detection results of LOFAR and BP-LOFAR are shown in Figure 10.

Figure 10 shows that both LOFAR and LOFAR-BP determine false alarms at the start and end and true alarms at the middle. To distinguish the two detection results of LOFAR and LOFAR-BP in detail, several statistics are listed, as shown in Table 3.

Table 3. Statistics for target detection.

Method	First Occurring Moment (s)	Last Vanishing Moment (s)	Judgment True Ratio
LOFAR	288.50	706.50	43.45%
LOFAR-BP	282.50	710.50	44.51%

In Table 3, the values of the first occurring moment indicate that LOFAR-BP detects the target 6 s earlier than LOFAR does. The values of the last vanishing moment indicate that LOFAR-BP loses the target 3.5 s later than LOFAR does. The values of the judgment true ratio indicate that, compared with LOFAR, LOFAR-BP improves the detection probability by 1.06%. Additionally, in Figure 6, at the detection time 282.50 s, the target signal is completely drowned in the ocean environment noise, which means that LOFAR-BP works

in a lower SNR ocean environment than LOFAR does. The results reveal that LOFAR-BP has a better target detection ability than LOFAR does.

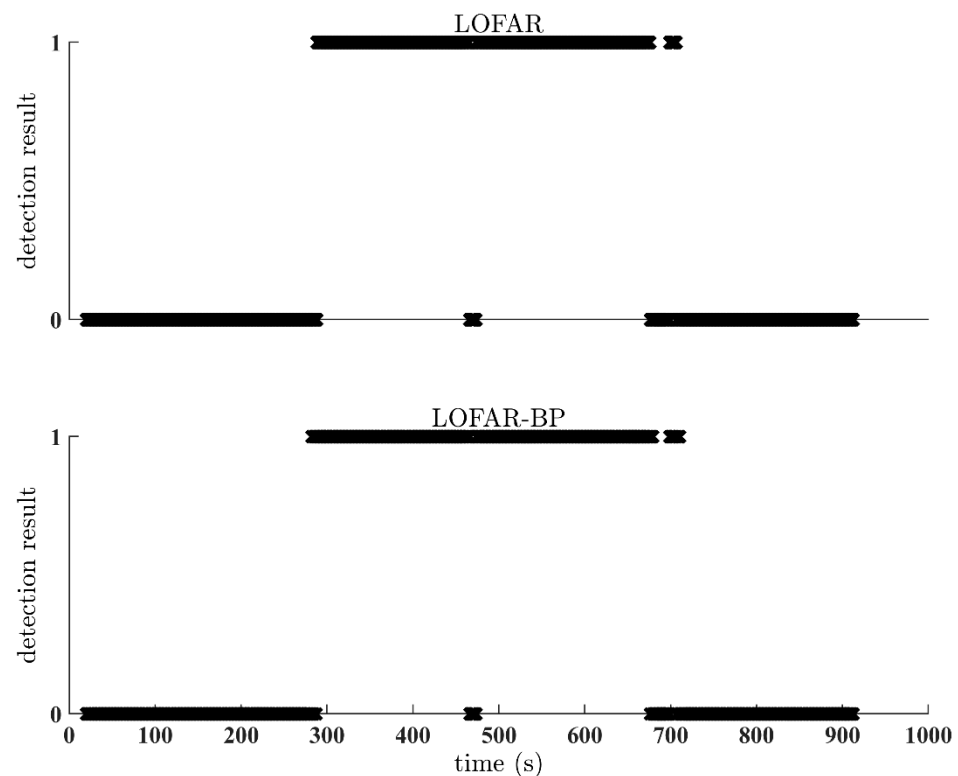


Figure 10. Detection result.

5. Conclusions

In this paper, a LOFAR-BP method was proposed. The method retains all the steps of LOFAR but replaces the FT step with a frequency analysis tool that is based on BP. Owing to the sparsity of BP, LOFAR-BP can collect signal energy into a small time–frequency area, with the goal of improving the frequency resolution. Both the simulation experiment and ocean experiment revealed that, compared with LOFAR, LOFAR-BP not only highlights the target signal and suppresses the noise better but also has a higher frequency resolution and SNR, making it detect the target earlier and track the target longer.

LOFAR-BP is based on the BP. Although the BP is a linear constrained programming problem, solving the BP requires a large time cost, even though obtaining a suboptimal solution. Owing to this, LOFAR-BP may not be suitable for real-time analysis. However, the adaptive line spectrum enhancer (ALE) can be used to improve the input SNR and offer the frequency of useful signals, which is a potential benefit to design a very small dictionary in BP, and then decrease the time complexity of LOFAR-BP. Although LOFAR-BP can improve the frequency resolution, whether the improvement ability has a limitation remains ambiguous. In LOFAR-BP, the parameter of the extended length that determines the frequency interval influences the frequency resolution indirectly. A larger extended length may result in a higher frequency resolution but at a higher computational cost. In the future, we will explore how the frequency resolution changes as this parameter increases and seek a proper value that balances the frequency resolution and computational cost in LOFAR-BP.

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Appendix A

Theorem A1. A signal is denoted by $y[n]$ $0 \leq n \leq N - 1$, an $N \times 1$ column vector. y is extended into $y_e = \begin{pmatrix} y \\ 0 \end{pmatrix}$ by adding zeros into a $K \times 1$ column vector, where K is the extended length larger than N . The FT result of y_e is denoted by $\hat{y}_e$, which is calculated using the following formula:

$$\hat{y}_e[n] = \sum_{i=0}^{i=N-1} y[i] \exp\left(j(-1)^n \frac{2\pi}{K} i\right) \quad 0 \leq n \leq K - 1 \quad (\text{A1})$$

We prove that $\hat{y}_e$ is the same as the optimal solution of the L2 norm constrained programming problem listed below:

$$\begin{aligned} \min \quad & \|x\|_2 \\ \text{s.t.} \quad & \Phi x = y \end{aligned} \quad (\text{A2})$$

where

$$\Phi = a_{r,c}|_{N \times K} \quad a_{r,c} = \frac{1}{K} \exp\left(jc \frac{2\pi}{K} r\right) \quad 0 \leq r \leq N - 1, \quad 0 \leq c \leq K - 1 \quad (\text{A3})$$

Proof. F is set as an $K \times K$ matrix below.

$$F = a_{r,c}|_{K \times K} \quad a_{r,c} = \frac{1}{\sqrt{K}} \exp\left(jc \frac{2\pi}{K} r\right) \quad 0 \leq r \leq K - 1, \quad 0 \leq c \leq K - 1 \quad (\text{A4})$$

F is an FT matrix that is a unitary matrix; thus, $F^{-1} = F^H$. F is divided into two matrices $F = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$. F_1 and F_2 are the first N rows in F and the last $K - N$ rows in F , respectively. $F^{-1} = F^H = \begin{pmatrix} F_1^H & F_2^H \end{pmatrix}$. F^{-1} is expanded below.

$$F^{-1} = \frac{1}{\sqrt{K}} \begin{pmatrix} \exp(j(-1)0\frac{2\pi}{K}0) & \exp(j(-1)0\frac{2\pi}{K}1) & \dots & \exp(j(-1)0\frac{2\pi}{K}(K-1)) \\ \exp(j(-1)1\frac{2\pi}{K}0) & \exp(j(-1)1\frac{2\pi}{K}1) & \dots & \exp(j(-1)1\frac{2\pi}{K}(K-1)) \\ \dots & \dots & \dots & \dots \\ \exp(j(-1)(K-1)\frac{2\pi}{K}0) & \exp(j(-1)(K-1)\frac{2\pi}{K}1) & \dots & \exp(j(-1)(K-1)\frac{2\pi}{K}(K-1)) \end{pmatrix} \quad (A5)$$

According to Formulas (11) and (15), and the matrix multiplication, the following conclusion is evident.

$$\hat{y}_e = \sqrt{K}F^{-1}y_e = \sqrt{K} \begin{pmatrix} F_1^H & F_2^H \end{pmatrix} \begin{pmatrix} y \\ 0 \end{pmatrix} = \sqrt{K}F_1^H y \quad (A6)$$

Because F is a unitary matrix, $F_1F_1^H = I$; $F_1F_2^H = O$; $F_2F_1^H = O$; $F_2F_2^H = I$; and the null space of F_1 is F_2^H .

According to Formula (13) and (14), and the way how to divide F into F_1 and F_2 , Φ is exactly $\frac{1}{\sqrt{K}}F_1$. For $\Phi x = y$, we obtain a particular solution $x = \sqrt{K}F_1^H y$. According to the particular solution and null space, the general solution for $\Phi x = y$ is $F_2^H \xi + \sqrt{K}F_1^H y$, where ξ is an arbitrary $(K - N) \times 1$ column vector. The L2 norm-constrained programming problem (Formula (12)) can be written as a least mean square below.

$$\min \left\| F_2^H \xi + \sqrt{K}F_1^H y \right\|_2 \quad (A7)$$

The two optimum solutions for Formulas (12) and (17) are denoted by x^* and ξ^* , respectively, and $x^* = F_2^H \xi^* + \sqrt{K}F_1^H y$.

The least mean square (Formula (17)) is solved. ξ^* and x^* are obtained below.

$$\begin{aligned} \xi^* &= \left((F_2^H)^H F_2^H \right)^{-1} (F_2^H)^H (-1) \sqrt{K}F_1^H y = O \\ x^* &= F_2^H \xi^* + \sqrt{K}F_1^H y = \sqrt{K}F_1^H y \end{aligned} \quad (A8)$$

Formulas (16) and (18) prove that Theorem 1 is true. $\square$

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